

document frequency for a given term (a logarithm of a reciprocal of what proportion of documents the term appears in).

For example, imagine that we are creating TFIDF vectors for biographies of US presidents. In the context of creating TFIDF vectors, we'll refer to each biography as a document. In the biography of Abraham Lincoln, the word *representative* will probably appear at least once, since he served in the Illinois House of Representatives and the US House of Representatives. If *representative* appears three times in the biography, then we say its term frequency is 3. More than a dozen presidents have served in the US House of Representatives, so maybe about 20 out of 44 total presidential biographies contain the term *representative*. We can then calculate the inverse document frequency as:

$$\log\left(\frac{44}{20}\right) = 0.788$$

The final value we're looking for is the term frequency times the inverse document frequency: $3 \times 0.788 = 2.365$. Now consider the term *Gettysburg*. It may appear twice in Lincoln's biography but never in any other, so the term frequency will be 2 and the inverse document frequency will be the following:

$$\log\left(\frac{44}{1}\right) = 3.784$$

The vector element associated with *Gettysburg* will be the term frequency times the inverse document frequency, which is $2 \times 3.784 = 7.568$. The TFIDF value for each term should reflect its importance in a document. Soon, this will be important for our chatbot's ability to determine user intent.

We don't have to calculate TFIDF manually. We can use a function from the `scikit-learn` module:

```
vctrz = TfidfVectorizer(ngram_range = (1, 1), tokenizer = normalize, stop_words = 'english')
```
